

MTE 203 – Advanced Calculus

Week 6 Homework – Solutions

Tangent Lines and Tangent Planes (S.12.9.)

Problem 1: [12.9, Prob. 19]

Find equations for the tangent line to the curve

$$x = t^2, \quad y = t, \quad z = \sqrt{t + t^4}$$

at the point $(1, 1, \sqrt{2})$

Solution:

At the point $(1, 1, \sqrt{2})$, we see that the value of $t = 1$. Hence, the direction of the line at the point $(1, 1, \sqrt{2})$ is given by,

$$\left. \frac{d\vec{r}}{dt} \right|_{t=1} = \left(2t, 1, \frac{1 + 4t^3}{2\sqrt{t + t^4}} \right) \Big|_{t=1}$$

$$\left. \frac{d\vec{r}}{dt} \right|_{t=1} = (2, 1, \frac{5}{2\sqrt{2}})$$

Since $(8, 4, 5\sqrt{2})$ is also the direction of a vector tangent to the line at the point $(1, 1, \sqrt{2})$, the parametric equations for the tangent line are given by,

$$x = 1 + 8u$$

$$y = 1 + 4u$$

$$z = \sqrt{2} + 5\sqrt{2}u$$

Problem 2: [12.9, Prob. 25]

Find an equation for the tangent plane to the surface

$$x = y \sin\left(\frac{\pi z}{2}\right)$$

at the point $(-1, -1, 1)$

Solution:

Re-writing the equation as a surface level,

$$f(x, y, z) = y \sin \frac{\pi z}{2} - x$$

The normal to the tangent plane to this surface at the point $(-1, -1, 1)$ is the $\vec{\nabla} f$. Hence,

$$\nabla(y \sin(\pi z/2) - x)|_{(-1, -1, 1)} = (-1, \sin(\pi z/2), (\pi y/2) \cos(\pi z/2))|_{(-1, -1, 1)} = (-1, 1, 0),$$

as is $(1, -1, 0)$, the equation of the tangent plane is $0 = (1, -1, 0) \cdot (x + 1, y + 1, z - 1) = x - y$.

Problem 3: [12.9, Prob. 31]

Find the directional derivative for the function

$$f(x, y, z) = xyz + xy + xz + yz$$

at $(1, -2, 5)$ perpendicular to the surface

$$z = x^2 + y^2$$

Solution:

Let us write the surface $z = x^2 + y^2$ as a function of three variables,

$$x^2 + y^2 - z = 0$$

A vector perpendicular to the surface at the point $(1, -2, 5)$, is the gradient to the surface at the given point,

$$\mathbf{n} = \nabla(x^2 + y^2 - z)|_{(1, -2, 5)} = (2x, 2y, -1)|_{(1, -2, 5)} = (2, -4, -1),$$

We want to find,

$$\pm \mathcal{D}_{\hat{n}} f|_{(1, -2, 5)} = \pm \vec{\nabla} f|_{(1, -2, 5)} \cdot \hat{n}$$

The gradient of the function $f(x, y, z) = xyz + xy + xz + yz$ at $(1, -2, 5)$ is,

$$\pm \vec{\nabla} f|_{(1, -2, 5)} = (yz + y + z, xz + x + z, xy + x + y)|_{(1, -2, 5)}$$

$$\pm \vec{\nabla} f|_{(1, -2, 5)} = \pm(-7, 11, -3)$$

The directional derivative is then,

$$\pm \mathcal{D}_{\hat{n}} f|_{(1, -2, 5)} = \pm(-7, 11, -3) \cdot \frac{(2, -4, -1)}{\sqrt{21}} = \pm \frac{55}{\sqrt{21}}$$

Problem 4: [12.9, Prob. 39]

Show that the sum of the intercepts on the x -, y -, and z -axes of the tangent plane to the surface $\sqrt{x} + \sqrt{y} + \sqrt{z} = \sqrt{a}$ at any point is a .

Solution:

Since a normal vector to the surface is $\nabla(\sqrt{x} + \sqrt{y} + \sqrt{z} - \sqrt{a}) = \left(\frac{1}{2\sqrt{x}}, \frac{1}{2\sqrt{y}}, \frac{1}{2\sqrt{z}}\right)$, the equation of the tangent plane at any point (x_0, y_0, z_0) is

$$0 = \left(\frac{1}{\sqrt{x_0}}, \frac{1}{\sqrt{y_0}}, \frac{1}{\sqrt{z_0}}\right) \cdot (x - x_0, y - y_0, z - z_0) = \frac{1}{\sqrt{x_0}}(x - x_0) + \frac{1}{\sqrt{y_0}}(y - y_0) + \frac{1}{\sqrt{z_0}}(z - z_0).$$

The x -intercept of this plane is given by

$$0 = \frac{1}{\sqrt{x_0}}(x - x_0) - \sqrt{y_0} - \sqrt{z_0} \implies x = x_0 + \sqrt{x_0}(\sqrt{y_0} + \sqrt{z_0}) = \sqrt{x_0}(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0}).$$

The y -intercept is given by,

$$0 = -\sqrt{x_0} + \frac{1}{\sqrt{y_0}}(y - y_0) - \sqrt{z_0}$$

$$y = \sqrt{y_0}(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0})$$

With a similar expression for the z -intercept, their sum is

$$\begin{aligned} \sqrt{x_0}(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0}) + \sqrt{y_0}(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0}) + \sqrt{z_0}(\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0}) \\ = (\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0})^2 = a. \end{aligned}$$

More practice problems¹

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

Warm-Up Problems

1. S. 12.9, Probs. 2, 10, 22`

Extra Practice Problems

1. S. 12.9, Probs. 34, 36

¹ You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).

Extra Challenge Problem

1. S. 12.9, Probs. 38